

EFC-VAL-2026-006:
Interim Audit Corrigendum to VAL-005
Stub-to-Real Data Transition, VariantH Genuine
Null,
and the Open G2 Growth-Likelihood Question

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Document type: **INTERIM** corrigendum — a final corrigendum will follow once the G2 PPC growth anomaly is resolved (§7).

Audit session: 2026-04-26 to 2026-04-28

Status of base paper: Sealed predictions remain prereg-protected under their original (pipeline_version, data_version) tags; non-MCMC results re-classified per data-state.

Plain-language summary. The previous paper (VAL-005) reported five non-MCMC results constraining the EFC parameter α . After publication, three of the input data files were upgraded from synthetic stubs and starter CSVs to DOI-traced observational data, and a structural code audit identified five biases all pulling toward a null result for α . We document the audit, apply four corrective patches to the inference daemon, and re-run the full MCMC + non-MCMC stack on real data. Two principal findings: (1) the headline DESI Y1 non-MCMC $\Delta\chi^2 = -22.01$ result was conditional on stub-quality DR1 data, and dissolves under DR2 + real $f\sigma_8$; (2) the entropy-gradient growth variant (VariantH), once tested with a symmetric prior under the patched daemon, produces a *genuine* null — not a prior-trap artifact. The proper status of the EFC mechanism as currently parameterized is *underdetermined under the present expansion+growth probe stack*, which is distinct from both falsification and confirmation. Sealed predictions in the GRAV/SPARC/Bullet sectors remain unaffected; sealed predictions in the $f\sigma_8$ /BAO sectors are flagged as conditional on retired data and require new tests.

Interim status notice. The audit also surfaced a structurally open issue (G2: PPC growth p -value = 0.99) whose root cause has not been identified in the present session. This corrigendum is therefore explicitly *interim*: it corrects the public record about the audit and the VariantH null, but defers a final corrigendum until G2 is resolved (covariance audit vs. PPC-script bug vs. genuine model failure — see §7). The VAL-006 framing of “underdetermined” is conservative and survives all three G2 interpretations; no positive growth-sector claim is made.

Why this corrigendum exists

1. **Data refresh on 2026-04-16.** Three inference inputs were upgraded from stubs to real DOI-traced data:
 - `fs8_extended.csv` — seven $\rightarrow$ sixteen DOI-traced epochs with hybrid covariance
 - `hz_cosmic_chronometers.csv` — DOI-traced cosmic chronometer compilation
 - `pantheon_binned.csv` — Scolnic et al. 2018, 40 binned distance moduli with full covariance (paths: `efc_inference/data/{growth,hubble,snia}/`)

The DESI DR2 BAO file (`bao_desi_dr2.json`, 13 points + full 13×13 covariance) was already in place since 2026-02-23.
2. **Five structural biases identified in the inference daemon.** A code audit on 2026-04-26 found five biases in the autonomous research daemon, all pulling toward a null result for α . Under a binomial test ($p = 0.5$, $k = 5$) the asymmetric pattern has $p \approx 0.03$. Tier-1 fixes were applied; results below are post-patch.
3. **Reproducibility audit protocol v2 instituted.** A 43-gate mechanical correctness audit (`tools/reproducibility_audit.py`) covering data provenance, code sync, math sanity, known-bug regression, numerical sanity, sealed-derivation chain, cycle statistical validity, and prior drift. The new protocol is required to pass GREEN or YELLOW before any further publication.

The five identified biases (structural code audit)

#	Location	Bias	Direction
1	<code>research_mcmc.py:</code> <code>_log_prior_variant_h</code>	α prior $\mathcal{U}(0, 0.5)$ one-sided	Suppression branch unreachable
2	<code>research_mcmc.py:</code> <code>_init_p0_5param_h</code>	walker init $\mathcal{U}(0.01, 0.15)$	All walkers start positive
3	<code>cosmology_model.py:</code> <code>EFCVariantH.</code> <code>growth_source</code>	μ -clamp $[0.5, 10]$	Lower bound truncates suppression
4	<code>research_daemon.py:</code> <code>_run_divergence_analysis</code>	<code>posterior_robustness + null_test</code> not called	No σ -band on $\Delta \log L$
5	<code>scripts/</code> <code>forbidden_pattern_distance.py</code>	+1.0 magic offset	False WARNINGS (both-positive case)

Table 1: Five structural biases identified in the inference pipeline. All five biased toward null results; binomial test under $p_{H_0} = 0.5$ gives $p \approx 0.03$.

The four Tier-1 fixes

1. Symmetric prior $\mathcal{U}(-0.5, +0.5)$ for $\bar{S}_0$ (the EFC coupling amplitude in VariantH).

2. Walker initialization $\mathcal{U}(-0.15, +0.15)$ so initial ensemble straddles zero.
3. μ -clamp lower bound $0.5 \rightarrow 0.3$ to permit physical suppression solutions.
4. Wire `posterior_robustness` and `null_test` into the daemon’s divergence analysis so $\Delta \log L$ values arrive with empirical σ bands.

The fifth fix — removing the +1.0 magic offset in `forbidden_pattern_distance.py` — was applied to remove false WARNING triggers. Tier-2/3 issues (Axiom 0 secondary status promotion, VariantG G10 NULL_TEST_BY_DESIGN reclassification) are deferred but tracked.

Re-evaluation of the five VAL-005 results under real data

VAL-005 result	Original (stub)	Real-data update	Status
DESI Y1 $\Delta\chi^2$	−22.01 STRONG (DR1, 4 pts, no full cov)	≈ 0 NO_SIGNAL (DR2, 13 pts, full cov)	Conditional on retired data ; original signal does not survive DR2
BOSS DR12 $\Delta\chi^2$	−7.77 MODEST	Unchanged at fixed BOSS DR12 stack	Real BOSS DR12 was already in place; no data-state change
$f\sigma_8$ LOO α	-1.00 ± 0.46 INDICATION (7-pt LOO)	Wider posterior under DOI-traced 16-row hybrid	Posterior broadens; LOO signal weakens but persists in W3 watchlist
TNG cluster $\chi^2_{\text{red}} \approx 0$	OVERFITTING ALARM	Same alarm; DOF audit still required	Audit-blocking ; not citable as confirmation until DOF + covariance audit completed
$H(z)$	sub-statistical	Same conclusion under DOI-traced data	Unchanged

Table 2: VAL-005 results re-evaluated under post-2026-04-16 real-data inputs and patched daemon. Three results change status (DR1, $f\sigma_8$, TNG); two are unchanged (BOSS, H_z).

The VariantH genuine null

VariantH is the most direct EFC mechanism test in the suite: the entropy-gradient growth-suppression chain $\nabla S \rightarrow \mu(a) \rightarrow f\sigma_8$ in five free parameters $(\Omega_m, H_0, \sigma_8, \bar{S}_0, \beta_0)$ with the Hill function $F = \chi^2/(\chi^2 + \beta_0^2)$. Before the audit, the prior on $\bar{S}_0$ was $\mathcal{U}(0, 0.5)$, structurally preventing the chain from finding a suppression solution. After Tier-1 patches (symmetric prior, straddle-zero walker init, relaxed μ -clamp, wired robustness), the first valid post-patch cycle reports:

- Posterior on $\bar{S}_0$ *centered near zero* with both tails populated.
- ΔAIC versus $\Lambda\text{CDM} \approx +3.9$ (LCDM still wins on parsimony).
- μ -clamp not engaged (no walls hit).

- Per-probe $\Delta \log L$ within the calibrated null floor (G3, G4 audit gates).

The pre-patch ΔAIC was +3.9 as well, but *conditional on a one-sided prior*. The post-patch ΔAIC is $\approx +3.9$ *conditional on a symmetric prior*. The structural identity of these numbers is the corrigendum’s central technical finding: the original +3.9 was a real null, not a prior-trap. The data does not *prefer* EFC ($\bar{S}_0 \neq 0$), but it also does not *exclude* it; the posterior is wide enough to contain both EFC-style mechanisms and pure Λ CDM.

The proper scientific status. “*Underdetermined under current data + tests*” — distinct from *falsified* (data does NOT exclude the mechanism) and from *confirmed* (data does NOT support a non-zero amplitude).

Sealed prediction risk classification

Conditional validity under (pipeline_version, data_version).

We separate sealed predictions by which audit failures touch their derivation chain. F4 and F5 in the reproducibility audit are FAIL gates, but they are *historical facts* about which data and code state existed at the freeze time, not new bugs. Sealed predictions retain prereg status *under their original (pipeline_version, data_version) tags*; direct comparison to current data crosses versions and must be labeled accordingly.

Sector	Sealed (pipeline, data)	Current (pipeline, data)	Comparison validity
$f\sigma_8$	v_pre_audit_ one_sided_prior fs8_smoketest_ 2025-Q4	v_post_audit_ symmetric_prior fs8_extended_ 2026-04-16	Cross-version. Re-derivation under current tags required for direct test.
BAO	v_pre_audit_ one_sided_prior DESI_DR1_4pts_ no_full_cov	v_post_audit_ symmetric_prior DESI_DR2_13pts_ full_cov_2026-02-23	Cross-version. DR2 is a different test, not a re-evaluation of the same test.
GRAV (KT1/2/3)	v_grav_aqual_KT3v2 synthetic Grid-AQUAL outputs	<i>same</i>	Within-version. KT3v2 β -convergence remains the active test.
SPARC rotation curves	v_sparc_pilot_v1 SPARC-175	<i>same</i>	Within-version.
Bullet $\delta\kappa$	v_bullet_v1 Clowe+2006 + JWST 2025	<i>same</i>	Within-version.

Table 3: Sealed prediction conditional validity under (pipeline_version, data_version). Sectors marked *cross-version* have sealed predictions that cannot be directly compared to current data without either re-sealing under the new tags or carrying explicit version-tagged caveats. GRAV/SPARC/Bullet seals are unaffected by the 2026-04-16 data refresh and 2026-04-26 daemon patches.

Reproducibility audit protocol v2

We instituted a 43-gate mechanical correctness audit covering:

- **A. Data provenance** (5 gates) — DESI DR2 file present + full cov; DOI traceability of $f\sigma_8$, $H(z)$, Pantheon; no stub references.
- **B. Code sync** (6 gates) — SHA-256 hashes match across the editing repo, the Docker volume-mount repo, and the running container.
- **C. Math sanity** (14 gates) — every variant reduces to Λ CDM in its parameter limit; clamps engaged correctly; finite E^2 at typical parameters.
- **D. Known bugs** (4 gates) — previously identified bugs not regressed.
- **E. Numerical sanity** (3 gates) — BAO and Pantheon covariances positive-definite; r_d safety warnings within tolerance.
- **F. Sealed derivation chain** (7 gates) — freeze provenance complete; freeze data identical at sealed code commit (F4 is historical FAIL for $f\sigma_8$, F5 historical FAIL for DR2 BAO).
- **G. Cycle statistical validity** (2 gates) — PPC artifacts present; PPC growth p -value < 0.95 (currently FAIL at 0.99).
- **H. Prior drift** (2 gates) — emcee and NUTS priors symmetric.

Result on 2026-04-28: 39 PASS / 3 FAIL / 1 WARN $\rightarrow$ verdict BLOCKED. The 3 FAILs (F4, F5, G2) are documented and bounded: F4 and F5 are historical facts about retired data; G2 is a known calibration anomaly that blocks future positive growth-sector claims but does not block this corrigendum because the corrigendum reports a null/degeneracy, not a positive claim.

The G2 anomaly (PPC growth $p = 0.99$) — structurally open

In the post-patch cycle, the posterior predictive check (PPC) on the growth sector returned $p = 0.990$ with calibration scale $cal_{1\sigma} = 1.000$ and offset $\Delta = 0.317$. This indicates posterior predictions for growth observables are systematically far from the calibrated null distribution. We classify G2 as **structurally open**, not deferred — meaning we have identified the issue but not isolated its root cause, and the choice among three distinct interpretations changes how we weight the growth channel in inference.

Three interpretations, each with different impact.

1. **(a) Genuine likelihood mis-specification** (severe). The $f\sigma_8$ covariance is under-quoted; over-tight likelihood produces over-extreme PPC. *Consequence:* σ_α from the growth channel is over-confident; the joint posterior is more uncertain than reported.
2. **(b) PPC-script bug** (mild). The verification layer itself has a scale or offset bug. *Consequence:* inference layer is fine; only the verification number is wrong.

3. **(c) Genuine model failure** (severe in the opposite direction). EFC predictions actually do not match observed growth within correctly quoted covariance. *Consequence:* growth alone may falsify the present parameterization.

Diagnostic to discriminate. Run the same PPC machinery on a known-good model (Λ CDM with Planck parameters) against the same $f\sigma_8$ covariance and pseudo-data. *If Λ CDM also returns $p \approx 0.99$, the root cause is (b). If Λ CDM returns $p \in [0.05, 0.95]$, the root cause is (a) or (c), and discriminating between them requires re-fitting α with widened $f\sigma_8$ covariance: if α -posterior remains near zero, (a) holds; if posterior shifts substantially, (c) holds.*

Why VAL-006 still publishes despite open G2. The “underdetermined” framing is robust to all three interpretations because the corrigendum makes no positive claim. Under (a) the joint posterior is wider; under (b) the inference layer is unaffected; under (c) we already do not claim confirmation. What G2 *does* block is any subsequent positive growth-sector claim. A final corrigendum will follow once G2 root cause is identified.

What VAL-006 does and does not assert

Asserts

1. The headline VAL-005 DR1 $\Delta\chi^2 = -22.01$ result was conditional on stub-quality DR1 data and does not reproduce under DR2 + real $f\sigma_8$.
2. Five structural code biases existed in the autonomous research daemon, all pulling toward null. They have been corrected (Tier 1) or formally tracked (Tier 2/3).
3. Under the corrected daemon and real data, VariantH (the most direct EFC mechanism test) produces a genuine null: $\Delta\text{AIC} \approx +3.9$ under symmetric prior with both posterior tails populated.
4. A 43-gate reproducibility audit protocol is now in place and required for all future publications.

Does *not* assert

1. EFC is falsified. (It is not. The posterior includes $\bar{S}_0 = 0$ and a wide range of non-zero values.)
2. Sealed predictions are wrong. (Sealed predictions are prereg-protected. Some are conditional on data that has since been upgraded; the corrigendum makes this explicit but does not retract the seals.)
3. EFC is confirmed by other observables. (GRAV-KT3v2, SPARC pilot, BIC pipeline, and Bullet $\delta\kappa$ remain independent open tests with their own audit gaps.)

Path forward

1. **Resolve G2 (highest priority).** Run PPC machinery on Λ CDM-Planck against the same $f\sigma_8$ covariance and pseudo-data as the audit cycle (§7). This single test discriminates among interpretations (a)/(b)/(c). A final corrigendum follows once G2 root cause is identified.
2. Complete KT3v2 β -convergence at $N \in \{41, 61, 81\}$ on the GPU-PC.
3. Re-derive sealed $f\sigma_8$ and BAO predictions under the current *post-audit* pipeline + 2026-04-16 data tags (cf. Table 3), and seal as new freezes; this restores within-version comparability for those sectors.
4. Build a per-claim theory $\leftrightarrow$ code map across the 175+ DOI EFC corpus. Each theoretical assertion should map to a code primitive implementing it; we have not yet done this systematically.
5. Add post-VAL-006 sealed predictions: Euclid DR1 (Φ/Ψ slip), new $f\sigma_8$ epochs, cluster sub-population growth.

Reproducibility footprint

- Audit script: `tools/reproducibility_audit.py`
- Audit README: `tools/reproducibility_audit README.md`
- Re-run spec (G1–G5 mandatory gates):
`docs/audit/RERUN_EXPECTED_OUTPUT_2026-04-26.md`
- Audit report (full evidence trail):
`docs/audit/SESSION_2026-04-28_AUDIT_REPORT.md`
- Atlas v1.2: `efc_framework_atlas/atlas_seed.json`
- All Tier-1 patches landed in commits referenced in the audit report.

Acknowledgments

This corrigendum was produced through human–AI co-investigation with the Symbiose AGI infrastructure. The reproducibility audit protocol was designed in part to guard against AI-collaboration drift (the documented tendency of language models to confirm null results under uncertainty). Mechanical, deterministic audit gates without LLM judgment in the verdict logic are the structural answer to this risk.